DIP-CSE340 ASSIGNMENT 1
Deadline : 2/9/2025

Total Marks : 40 Marks **Total Days : 14 Days**

Instructions:

- Standard institute plagiarism policy holds.
- This is an individual assignment. State any assumptions you have made clearly.
- Even a second late will be considered late, so do submit your assignments at least before 11:56 PM on the date of the deadline to avoid any unexpected issues. Late submissions are not entertained.
- Viva 5 marks will be taken during the demos (written/oral).
- Submit a Jupiter notebook (dip_rollnumber.ipny) with results from section A, B and C. You can use either Python/Matlab. Jupiter Notebook for Matlab is also available. Wherever required use text cell/comment to answer the questions.

Section A - Pixels and storage

(CO1&2)(12.5 marks)

A high-definition television (HDTV) system generates images with 1125 horizontal TV lines interlaced (i.e., every other line is painted in two fields, each field lasting 1/60th of a second). The aspect ratio is 16:9. The vertical resolution is therefore fixed at 1125 lines, while the horizontal resolution is proportional to the vertical resolution by the width-to-height ratio. Each pixel in the digital colour image has 24 bits (8 bits each for red, green, and blue). A company extracts digital images from a two-hour HDTV movie using this system.

1. Answer the following questions in the Notebook: (7 marks)
   (a) Calculate the horizontal resolution (pixels per line) of the frame based on the given aspect ratio.
   (b) What are the total number of pixels per frame?
   (c) Explain how pixel coordinates are conventionally represented in digital imaging (e.g., origin, axes directions).
   (d) Calculate the total number of bits required to store a single frame.
   (e) How many bits are required to store the entire two-hour HDTV movie? Express your result in Gigabytes (assume 1 GB $= 2^{30}$ bytes).
   (f) If the system down samples the image by a factor of 2 in both horizontal and vertical directions, what is the new frame resolution and storage requirement?
   (g) If the quantization is reduced from 24 bits/pixel to 12 bits/pixel, how much storage would be saved for the two-hour movie?

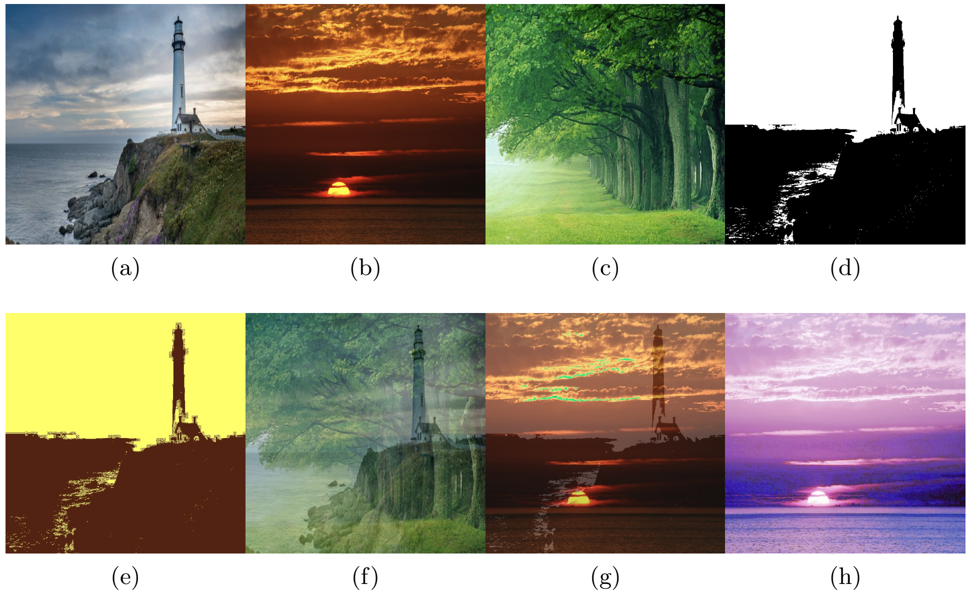

Figure 1: The images (a-d) are the Input images and (e-h) are the outputs generated using the the input images in (a-d). (d) is a binary mask generated from (a), convert the same to a 3-channel image by duplication (this implies the different channels R,B and G are same)

2. Write a program given an image file and a target aspect ratio (”16:9”) to computes:(5.5 marks)
   (a) The adjusted resolution of the image (width height in pixels) that fits the target aspect ratio using floor rounding mode,
   (b) The total number of pixels in the adjusted frame.

Section B - Operations on Image I

(CO1,2&3)(12.5 marks)

Consider the images in the Figure 1. The output images are generated using the function (A-D).

A. $g(I_1, I_2)$ B. $g(I_1, 255 - I_2)$ C. $f(I_1, I_2, 0.5, 0.5, 0)$ D. $f(I_1, I_2, 0.8, 0.2, 6)$

Here $f(I_1, I_2, k_1, k_2, k_3) = ((I_1 * k_1) + (I_2 * k_2) + k_3) \mod 255$ and $g =$ Histogram matching(I_1, I_2), where in histogram matching I_1 is the source and I_2 is the reference image respectively and $\mod$ denote modulo operator.

1. Identify the inputs images I_1 and I_2 from Figure 1 images (a-d) and match the functions used to generate the Figure 1 images (e-h). (For example Your answer should be as follows : Image e = function (A) $g(I_1, I_2)$ with $I_1 =$ image (a) and $I_2 =$ image (a)). (4 marks)
2. Write a program to generate the images (e-h) using their respective functions (A-D) in the Notebook. Also plot the histogram of each channels (R,G & B). (4 marks)
3. Write the algorithm for implementing histogram matching from scratch. (2.5 marks)

(a) Input (b) Input (c) Output

Figure 2: The images (a,b) are Binary grayscale mask extracted from the Figure 1 images(a,b).

4. Write a program to generate the mirror images of images (a-b) in Figure 1. (2 mark) Note: Only the images (a-c) are given. Extract the binary mask (d) from the image (a) by using binary thresholding . Modify the binary mask to make it resemble the given image in Figure 1(d).

Section C - Operations on Image II

(CO1,2&3)(10 marks)

Consider the Grayscale Binary images (a,b) in the Figure 2 with grayscale values 0&255.

1. Write a program to extract the Grayscale Binary images (a,b) Figure 2 from the images (a,b) Figure 1. The extracted images (a,b) in the Figure 2 must be visibly similar. Marks will not be awarded if in (i)image (a) Figure 2 top part of the lighthouse and the bottom part of the cliff are not as detailed as in the figure, i.e the image (d) Figure 1 will not be accepted. (ii) image (b) the sun and clouds should be present. Make sure to clear the objects in the white space as much as possible. Some minor differences are accepted. (3 marks)

2. Change the grayscale value 0 in the grayscale binary image (b) Figure 2 to 100 and denote it as $\hat{b}$. Plot the image $\hat{b}$. Write a program to generate the image (c) Figure 2 using the images (a) Figure 2 and the new image $\hat{b}$. Note : image (c) Figure 2 should contain lighthouse, cliff, sun and clouds. (3 marks)

3. Perform Histogram equalisation on the image (c) Figure 2. Compare the histograms and the images before and after equalisation. Comment on the outputs. Is there a better method for equalising the images? If yes demonstrate with this example. (0.5+0.5+1+2 marks)

References and Additional Instructions:

- Chapters 1,2, and 3 (up to convolution) from the DIP book (referenced in the classroom) and Open-CV documentation. You can use inbuilt functions and LLMs. But the output and code should be there in the notebook submitted. You are free to attach a pdf file of the notebook.
- The viva will also include Lecture notes till the date of the deadline and questions related to work done in the assignment.

- In the Algorithm, do not add statements with inbuilt functions such as "Use the cv2.cvtcolor function to covert RBG to greyscale". Such statements will result in -2 marks.

Any code with run time error during demo (in any section A, B and C) will result in 0 marks. Do not use application such as Microsoft paint. Images generated without code will get -5 marks.

End
